Title: Prediction of Soccer Match Result with Machine Learning

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Introduction

Soccer is one of the most unpredictable games in the world. This game is also considered a low-scoring game compared with many other sports. Across many leagues and competitions, the average total goals per match typically ranges between ~2.4 to ~2.7 goals/game [1]. On top of that, it also has a high rate of 'draw' outcome, which makes the prediction harder. Based on the last 10 seasons in the English Premier league about 23% of matches end in a draw, roughly one in four matches [2]. It also has factors like player injuries and ground variance. Due to this unpredictability, various bets are placed every year on it. Among all these bets, most of them are not calculated. That is why most people end up losing money. To solve this issue, I want to use Machine Learning to predict the match results.

The problem of predicting soccer match results has attracted increasing attention from researchers due to the sport's inherent uncertainty and the availability of large historical datasets. Traditional statistical approaches, such as Poisson and regression-based models, have been widely used to model goal scoring and match outcomes by assuming low event rates and independence between scoring events [4]. More recently, machine learning techniques have been applied to this problem to capture nonlinear relationships and interactions among multiple match-related factors [3]. Prior studies have explored methods such as logistic regression, decision trees, random forests, and gradient boosting to predict match outcomes or goal counts, often demonstrating improvements over simple baseline models [5]. However, despite these advance modeling, they couldn't accurately predict most of the matches. In addition, most of the work did not consider the drawing factor and low-scoring issues. This motivates further investigation into how machine learning models can be designed and evaluated specifically for soccer matches.

In this project, soccer match result prediction is formulated as a supervised machine learning classification problem, where historical match data are used to predict future outcomes. The proposed approach focuses on modeling match results using machine learning algorithms while explicitly accounting for the high frequency of draws and the low number of goals typically observed in soccer matches. The contribution of this work lies in systematically analyzing how these characteristics affect predictive performance and in comparing multiple machine learning models under a consistent evaluation framework. While other studies did not consider the minor factors, this study will consider those and aims to provide insight into the strengths and limitations of machine learning for soccer match prediction and to contribute to a clearer understanding of when such models can outperform the prediction strategies.

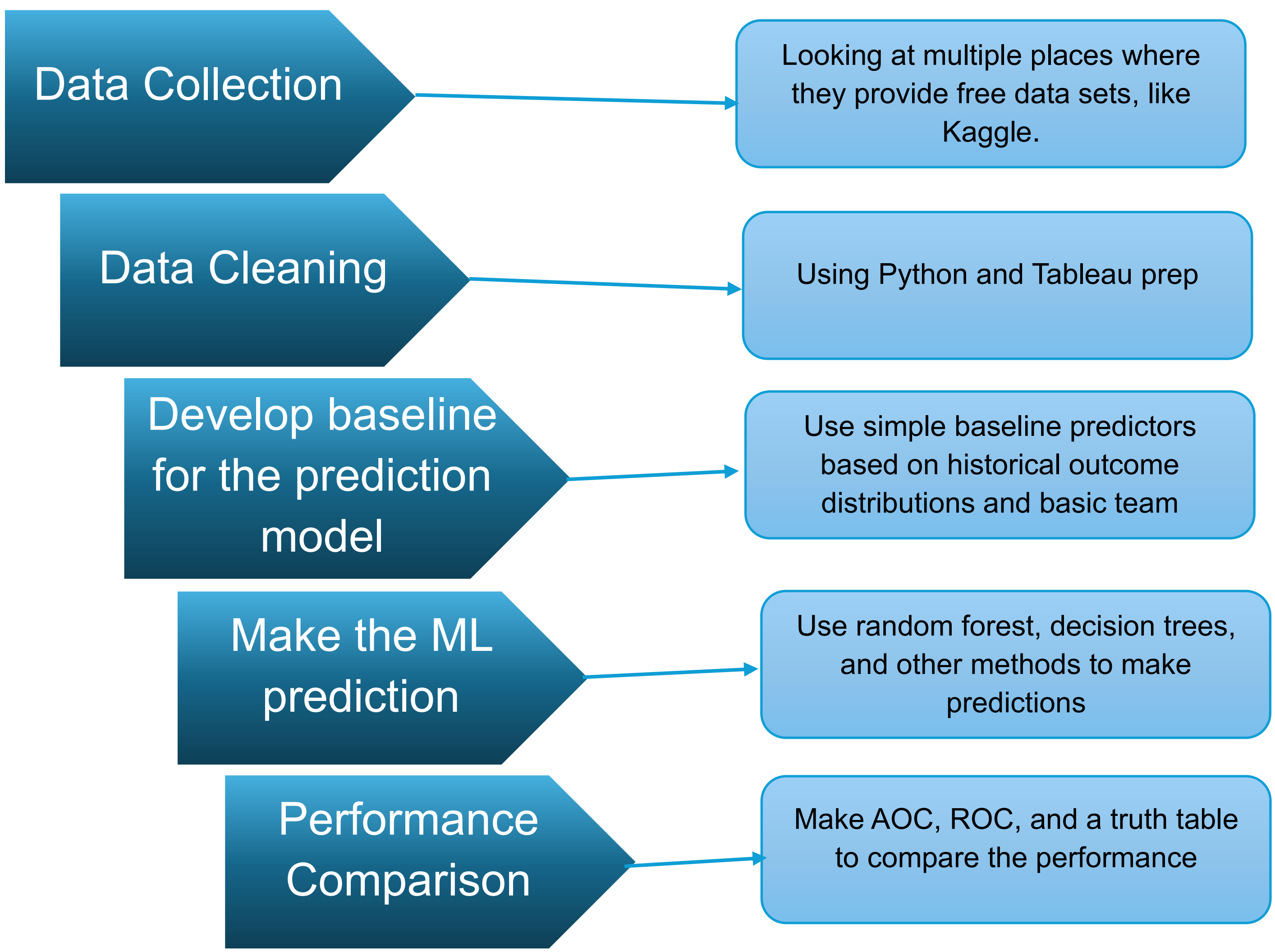
Hypothesis

I think the machine learning model can predict 15% more accurately than the baseline model based on historical outcome data.

Research Question

Can machine learning models predict soccer match outcomes more accurately than simple baseline models based on historical outcome data?

Methods: Research



Reference:

Reference 1: Dan. (2016, March 2). *Most soccer matches are within one goal of 1-0 | R-bloggers*. R-bloggers. https://www.r-bloggers.com/2016/03/most-soccer-matches-are-within-one-goal-of-1-0/

Reference 2: *Football Draws Stats | FootyStats*. (n.d.). Footystats.org. https://footystats.org/stats/draws

Reference 3: Usman Haruna, Jaafar Zubairu Maitama, Mohammed, M., & Ram Gopal Raj. (2022). Predicting the Outcomes of Football Matches Using Machine Learning Approach. *Communications in Computer and Information Science*, 92–104. https://doi.org/10.1007/978-3-030-95630-1_7

Reference 4: *Most soccer matches are within one goal of 1-0 | R-bloggers*. (2016, March 2). R-Bloggers. https://www.r-bloggers.com/2016/03/most-soccer-matches-are-within-one-goal-of-1-0/

Reference 5: Bunker, R., Yeung, C., & Fujii, K. (2024, March 12). *Machine Learning for Soccer Match Result Prediction*. ArXiv.org. https://doi.org/10.48550/arXiv.2403.07669